

DeltaFlow Labs Architecture

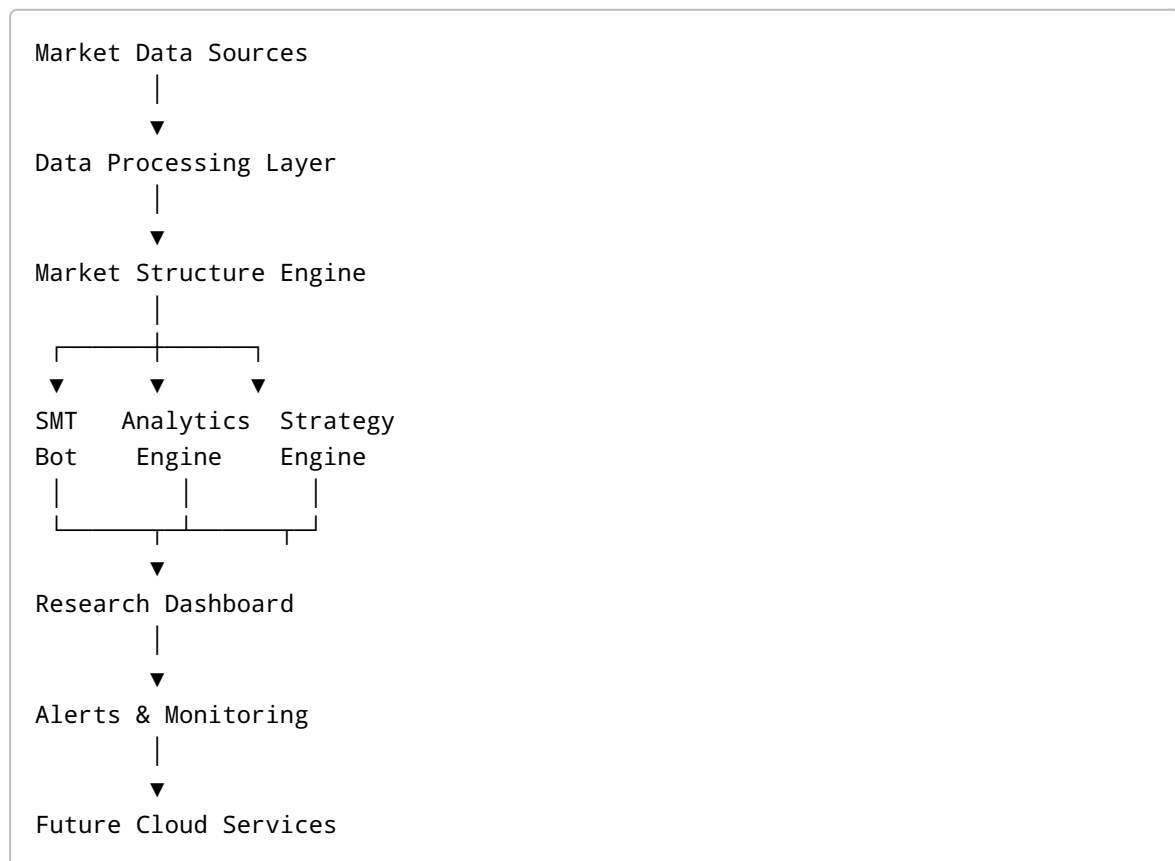
Overview

DeltaFlow Labs is designed as a modular market intelligence platform focused on quantitative research, market structure analytics, and automation infrastructure.

The architecture is built around independent research modules that communicate through a shared analytics and visualization layer.

This modular approach enables rapid experimentation, scalability, and future cloud-native deployment.

High-Level Architecture



Core Components

1. Market Data Layer

Purpose:

Collect and process real-time market data from supported platforms.

Supported Sources:

- MetaTrader 5
- Forex Markets
- Index Markets
- Cryptocurrency Markets
- Options Data Sources

Responsibilities:

- Price Collection
 - Candle Generation
 - Timeframe Management
 - Data Normalization
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2. Data Processing Layer

Purpose:

Transform raw market data into structured analytical information.

Responsibilities:

- Data Validation
- Candle Processing
- Historical Data Storage
- Event Generation
- Market State Updates

Outputs:

- Processed Candles
 - Market Events
 - Analytical Data Streams
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3. DeltaFlow Engine

Purpose:

Serve as the primary market structure intelligence engine.

Core Functions:

Structure Detection

- Swing High Detection

- Swing Low Detection
- Internal Structure Mapping
- External Structure Mapping

Structural Events

- Break of Structure (BOS)
- Change of Character (CHoCH)
- Trend Identification
- Liquidity Mapping

Market Context

- Bullish Structure Analysis
- Bearish Structure Analysis
- Consolidation Detection

Status:

Active Development

4. ICT SMT Bot

Purpose:

Research and detect SMT divergence opportunities across correlated markets.

Core Functions:

- SMT Divergence Detection
- Market Correlation Analysis
- Cross-Market Validation
- Signal Monitoring

Supported Examples:

- NIFTY vs BANK NIFTY
- BTC vs ETH
- Correlated Forex Pairs

Status:

Active Development

5. VEMB15 AUTO

Purpose:

Provide a framework for systematic strategy research and execution.

Core Functions:

- Signal Validation
- Risk Management Logic
- Position Management
- Strategy Monitoring

Status:

Research & Development

6. NIFTYOPT Framework

Purpose:

Monitor options market activity and support structured analytics.

Core Functions:

- Options Analysis
- Market Observation
- Alert Generation
- Dashboard Integration

Status:

Research & Development

Dashboard Layer

Technology:

- Streamlit

Purpose:

Provide a centralized visualization environment for all research modules.

Features:

- Real-Time Market Monitoring
- Structure Visualization
- SMT Monitoring
- Alert Dashboard
- Performance Analytics

Current Status:

Alert & Notification Layer

Purpose:

Deliver actionable events generated by research systems.

Supported Channels:

- Dashboard Notifications
- Telegram Alerts
- Future Web Notifications

Examples:

- BOS Detection
 - CHoCH Detection
 - SMT Divergence Events
 - Strategy Alerts
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Technology Stack

Programming

- Python

Trading Integration

- MetaTrader 5

Dashboard

- Streamlit

Web Platform

- HTML
- CSS
- JavaScript
- Cloudflare Pages

Cloud Infrastructure

- AWS (Planned)
- Cloudflare

Version Control

- Git
 - GitHub
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Security Principles

DeltaFlow Labs follows these security practices:

- No broker credentials stored in repositories
- No API keys committed to source control
- Environment variable based configuration
- Access separation between development and production systems

Sensitive information is excluded through repository management policies.

Future Architecture Goals

AI Market Intelligence Layer

Planned Features:

- Pattern Recognition
 - Market Classification
 - Research Assistance
 - Decision Support Systems
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Cloud-Native Platform

Planned Features:

- AWS Deployment
 - Scalable Processing
 - API Services
 - User Management
 - Enterprise Dashboard
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Design Principles

DeltaFlow Labs is built around:

1. Modularity

2. Scalability
 3. Transparency
 4. Research-Driven Development
 5. Real-Time Analytics
 6. Automation Infrastructure
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Current Development Focus

Primary development efforts are concentrated on:

- DeltaFlow Engine
 - ICT SMT Bot
 - Research Dashboard
 - Real-Time Market Processing
 - Market Structure Intelligence
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Last Updated: June 2026

DeltaFlow Labs

Quantitative Research • Market Intelligence • Automation Infrastructure